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Plate assembly and home appliance

Abstract

A refrigerator including a storage compartment and a cold air duct to guide cold air to the storage compartment, the cold air duct including a front plate facing the storage compartment, the front plate including a front body with a first coupling portion, a rear plate detachably coupled to the front plate, the rear plate including a rear body with a second coupling portion, and a slit in the rear body so that the second coupling portion is elastically deformable, and a heat insulating member between the front plate and the rear plate. The first coupling portion is coupled to the second coupling portion. The first coupling portion and the second coupling portion are decouplable by an external force pressing on the first coupling portion.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5779197	12/1997	Kim	N/A	N/A
10690390	12/2019	Kim et al.	N/A	N/A
2015/0132078	12/2014	Yamamoto	N/A	N/A
2023/0332819	12/2022	Park et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
216694137	12/2021	CN	N/A
217483056	12/2021	CN	N/A
4 047 292	12/2022	EP	N/A
10-135652	12/1997	JP	N/A
11-28985	12/1998	JP	N/A
2003-32862	12/2002	JP	N/A
2008-51126	12/2007	JP	N/A
5377954	12/2012	JP	N/A
5648756	12/2014	JP	N/A
5719485	12/2014	JP	N/A
2022070725	12/2021	JP	F25D 11/02
2022-529441	12/2021	JP	N/A
10-0333813	12/2001	KR	N/A
10-1011634	12/2010	KR	N/A
10-1844852	12/2017	KR	N/A
10-1852677	12/2017	KR	N/A
10-2022-0082388	12/2021	KR	N/A

OTHER PUBLICATIONS

English language translation of JP2022070725 to Hayakawa. Translated Apr. 2025. (Year: 2022). cited by examiner
International Search Report dated Dec. 13, 2023 International application No.

PCT/KR2023/013164. cited by applicant
Written Opinion of the International Searching Authority dated Dec. 13, 2023 International application No. PCT/KR2023/013164. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation application, under 35 U.S.C. § 111(a), of International Application No. PCT/KR2023/013164, filed on Sep. 4, 2023, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2022-0175305, filed on Dec. 14, 2022, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.

BACKGROUND
1. Field
(1) The disclosure relates to an improved plate assembly and a home appliance including the same.
2. Description of the Related Art
(2) A refrigerator is an appliance that stores food in a fresh state by including a main body having a storage compartment and a cold air supply system configured to supply cold air to the storage compartment. The storage compartment includes a refrigerating compartment maintained at a temperature of about 0 to 5° C. to refrigerate and store food and a freezing compartment maintained at a temperature of about −30 to 0° C. to freeze and store food. Generally, the storage compartment is provided to have an open front surface to allow food to be put in and taken out, and the open front surface of the storage compartment is opened and closed by a door.
(3) The refrigerator repeats a cooling cycle in which a refrigerant is compressed, condensed, expanded, and evaporated using a compressor, a condenser, an expander, and an evaporator. Here, both the freezing compartment and the refrigerator compartment may be cooled by a single evaporator provided at the freezing compartment side, or an evaporator may be provided in each of the freezing compartment and the refrigerator compartment for the freezing compartment and the refrigerator compartment to be independently cooled.
(4) Types of refrigerators may be classified according to the forms of the storage compartment and the door. For example, refrigerators may be classified into top mounted freezer (TMF) refrigerators in which a storage compartment is vertically divided by a horizontal partition and a freezing compartment is formed at an upper side and a refrigerator compartment is formed at a lower side, and bottom mounted freezer (BMF) refrigerators in which the refrigerator compartment is formed at the upper side and the freezing compartment is formed at the lower side.
SUMMARY
(5) Aspects of embodiments of the disclosure will be set forth in part in the description which follows and, in part, will be apparent from the description, or may be learned by practice of the presented embodiments.
(6) According to an embodiment of the disclosure, a refrigerator includes a storage compartment: an evaporator configured to generate cold air; and a cold air duct configured to guide the cold air generated in the evaporator to the storage compartment, the cold air duct including a front plate facing the storage compartment, the front plate including a front body, and a first coupling portion on a rear surface of the front body, a rear plate detachably coupled to a rear of the front plate, the rear plate including a rear body, a second coupling portion including an opening in the rear body,

and a locking portion, and a slit in the rear body configured so that the second coupling portion is elastically deformable, and a heat insulating member disposed between the front plate and the rear plate. The first coupling portion is coupled to the second coupling portion as a portion of the first coupling portion is inserted into the opening in the rear body, and the locking portion protrudes toward and contacts the portion of the first coupling portion inserted into the opening in the rear body. The first coupling portion and the second coupling portion are configured so that the first coupling portion and the second coupling portion are decouplable by an external force pressing on the first coupling portion.

(7) According to an embodiment of the disclosure, the first coupling portion may include a first coupling protrusion extending in a first direction directed toward the rear plate, a second coupling protrusion spaced apart from the first coupling protrusion and extending in the first direction, and a bridge connecting the first coupling protrusion and the second coupling protrusion, the bridge including a recessed portion recessed in a second direction opposite to the first direction.

(8) According to an embodiment of the disclosure, in response to the external force pressing on the recessed portion of the bridge in the second direction in a state in which the front plate and the rear plate are coupled to each other, the first coupling protrusion and the second coupling protrusion are moved toward one another so that the first coupling portion and the second coupling portion are decoupled.

(9) According to an embodiment of the disclosure, in response to the external force pressing the first coupling protrusion and the second coupling protrusion to be moved toward one another in a state in which the front plate and the rear plate are coupled to each other, the bridge is elastically deformed and the first coupling portion and the second coupling portion are decoupled.

(10) According to an embodiment of the disclosure, the refrigerator may further include a cap corresponding to the opening and detachably mounted on the opening to prevent moisture from entering the opening.

(11) According to an embodiment of the disclosure, the rear plate may include a cover configured to open and close the opening to prevent moisture from entering the opening.

(12) According to an embodiment of the disclosure, the slit may extend along a first direction in which the first coupling protrusion and the second coupling protrusion are arranged.

(13) According to an embodiment of the disclosure, the slit may include a first slit portion, and a second slit portion spaced apart from the first slit portion. The first coupling portion may be disposed between the first slit portion and the second slit portion in a state in which the front plate and the rear plate are coupled to each other.

(14) According to an embodiment of the disclosure, the second coupling portion may include a coupling body configured to surround the portion of the first coupling portion inserted into the opening in the rear body. The slit may include a first slit forming portion along a portion of an edge of the coupling body, a second slit forming portion extending in a straight line from a first side of the first slit forming portion, and a third slit forming portion extending in a straight line from a second side of the first slit forming portion.

(15) According to an embodiment of the disclosure, the slit may include a first slit forming portion spaced apart from the opening and extending in a direction perpendicular to a direction in which the first coupling protrusion and the second coupling protrusion are arranged, a second slit forming portion extending from a first side of the first slit forming portion toward the opening along the direction in which the first coupling protrusion and the second coupling protrusion are arranged, and a third slit forming portion extending from a second side of the first slit forming portion toward the opening along the direction in which the first coupling protrusion and the second coupling protrusion are arranged.

(16) According to an embodiment of the disclosure, the first coupling portion may include a first groove on the first coupling protrusion, and accessible through the opening from behind the rear body, and a second groove on the second coupling protrusion, and accessible through the opening

from behind the rear body.
(17) According to an embodiment of the disclosure, the bridge may include a first bridge portion extending from the first coupling protrusion to the recessed portion, and sloping toward the front body, and a second bridge portion extending from the second coupling protrusion to the recessed portion, and sloping toward the front body.
(18) According to an embodiment of the disclosure, the recessed portion may be provided at a center of the bridge.
(19) According to an embodiment of the disclosure, the heat insulating member may be configured to be in surface contact with at least one of the front plate and the rear plate.
(20) According to an embodiment of the disclosure, the heat insulating member may include a support hole corresponding to the opening of the second coupling portion and configured to accommodate the first coupling portion, and a seating portion extending inwardly from a rear surface of the heat insulating member about the support hole, and configured to support the locking portion of the second coupling portion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) These and/or other embodiments of the disclosure will become apparent and more readily appreciated from the following description of embodiments, taken in conjunction with the accompanying drawings of which:
(2) FIG. **1** is a perspective view illustrating an example of a refrigerator according to an embodiment of the disclosure.
(3) FIG. **2** is a cross-sectional side view illustrating an example of a refrigerator according to an embodiment of the disclosure.
(4) FIG. **3** is a perspective view illustrating an example of a plate assembly according to an embodiment of the disclosure.
(5) FIG. **4** is a perspective view illustrating the plate assembly shown in FIG. **3** when viewed from the rear.
(6) FIG. **5** is an exploded perspective view illustrating the plate assembly shown in FIG. **3**.
(7) FIG. **6** is an exploded perspective view illustrating the plate assembly shown in FIG. **5** from the rear.
(8) FIG. **7** is an enlarged view of a portion illustrating an example of a plate assembly according to an embodiment of the disclosure.
(9) FIG. **8** is a cut-away perspective view illustrating a state before a plate assembly according to an embodiment of the disclosure is assembled.
(10) FIG. **9** is a cutaway perspective view illustrating a state in which the plate assembly shown in FIG. **8** is assembled.
(11) FIG. **10** is a cross-sectional view illustrating a state in which a first plate and a second plate of a plate assembly according to an embodiment of the disclosure are coupled.
(12) FIG. **11** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **10** are being disassembled from each other.
(13) FIG. **12** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **11** are disassembled from each other.
(14) FIG. **13** is a cross-sectional view illustrating a state in which a first plate and a second plate of a plate assembly according to an embodiment of the disclosure are coupled.
(15) FIG. **14** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **13** are being disassembled from each other.
(16) FIG. **15** is a cross-sectional view illustrating a state in which the first plate and the second

plate of the plate assembly shown in FIG. **14** are disassembled from each other.
(17) FIG. **16** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment of the disclosure.
(18) FIG. **17** is a cross-sectional view of the plate assembly shown in FIG. **16**.
(19) FIG. **18** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **17** are being disassembled from each other.
(20) FIG. **19** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment of the disclosure.
(21) FIG. **20** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment of the disclosure.
(22) FIG. **21** is a view illustrating a state in which a cap shown in FIG. **20** is mounted on an opening according to an embodiment of the disclosure.
(23) FIG. **22** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment of the disclosure.
(24) FIG. **23** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment of the disclosure.
(25) FIG. **24** is a view illustrating a state in which the cover shown in FIG. **23** covers the opening.

DETAILED DESCRIPTION

(26) The various embodiments of the disclosure and terminology used herein are not intended to limit the technical features of the disclosure to the specific embodiments, but rather should be understood to cover all modifications, equivalents, and alternatives falling within the concept and scope of the disclosure.
(27) In the description of the drawings, like numbers refer to like elements throughout the description of the drawings.
(28) The singular forms preceded by “a,” “an,” and “the” corresponding to an item are intended to include the plural forms as well unless the context clearly indicates otherwise.
(29) In the disclosure, a phrase such as “A or B,” “at least one of A and B,” “at least one of A or B,” “A, B or C,” “at least one of A, B and C,” and “at least one of A, B, or C” may include any one of the items listed together in the corresponding phrase of the phrases, or any possible combination thereof.
(30) The term “and/or” includes combinations of one or all of a plurality of associated listed items.
(31) As used herein, such terms as “1st” and “2nd,” or “first” and “second” may be used to simply distinguish a corresponding component from another, and does not limit the components in other aspect (for example, importance or order).
(32) When one (e.g., a first) element is referred to as being “coupled” or “connected” to another (e.g., a second) element with or without the term “functionally” or “communicatively,” it means that the one element is connected to the other element directly, wirelessly, or via a third element.
(33) It will be understood that the terms “include”, “comprise” and/or “have” when used in this specification, specify the presence of stated features, integers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.
(34) It is to be understood that if a certain component is referred to as being “coupled to,” “coupled to,” “supported on” or “in contact with” another component, it means that the component may be coupled to the other component directly or indirectly via a third component.
(35) Throughout the specification, when a member is referred to as being “on” another member, the member is in contact with another member or yet another member is interposed between the two members.
(36) On the other hand, in this specification, the terms “front-rear direction,” “front,” “rear,” “upper-lower direction,” “height direction,” “upper side,” “lower side,” “left-right direction,” “left side,” “right side,” and the like are defined based on the drawings, but the terms may not restrict

the shape and position of the respective components.
(37) Embodiments of the disclosure may provide a plate assembly and a home appliance with an improved structure.
(38) Embodiments of the disclosure may provide a plate assembly and a home appliance capable of facilitating coupling and/or disassembling between parts.
(39) Embodiments of the disclosure may provide a plate assembly and a home appliance capable of one-touch assembly and/or one-touch disassembly.
(40) Embodiments of the disclosure may provide a plate assembly and a home appliance capable of preventing moisture from entering a coupling region between parts.
(41) Embodiments of the disclosure may provide a plate assembly and a home appliance capable of preventing a portion exposed to the outside from being sunk.
(42) Embodiments of the disclosure may provide a plate assembly and a home appliance with an improved sealing property.
(43) Embodiments of the disclosure are not limited to the various aspects mentioned above, and other aspects may become apparent to those of ordinary skill in the art based on the following descriptions.
(44) Hereinafter, embodiments according to the disclosure will be described in detail with reference to the accompanying drawings.
(45) FIG. **1** is a perspective view illustrating an example of a refrigerator according to an embodiment. FIG. **2** is a cross-sectional side view illustrating an example of a refrigerator according to an embodiment.
(46) Referring to FIGS. **1** and **2**, a refrigerator **1** will be described as an example of a home appliance including a plate assembly **80***a*. However, home appliances according to the disclosure are not limited to the refrigerator **1**. For example, the plate assembly **80***a* may be applied to various home appliances, such as a washing machine, a clothes care device, a shoe care device, an air conditioner, an air purifier, a humidifier, a cooking appliance, a dishwasher, a television (TV), and the like.
(47) Referring to FIGS. **1** and **2**, a refrigerator **1** may include a main body **10**. The refrigerator **1** may include a storage compartment **20** provided inside the main body **10**. The refrigerator **1** may include, a door **30** configured to open and close the storage compartment **20**. The refrigerator **1** may include a cooling system configured to supply cold air to the storage compartment **20**.
(48) The main body **10** may form at least a part of the external appearance of the refrigerator **1**. The main body **10** may have a front side that is open to allow a user to put food in the storage compartment **20** or take food out of the storage compartment **20**. The main body **10** may include an opening **10***a*. The opening **10***a* of the main body **10** may be opened and closed by the door **30**.
(49) The door **30** may be provided to open and close the main body **10**. The door **30** may be configured to open and close the opening **10***a* of the main body **10**. The door **30** may be rotatably coupled to the main body **10**. For example, the door **30** may be rotatably coupled to the main body **10** by a hinge **40** connected to each of the door **30** and the main body **10**.
(50) An outer surface **31** of the door **30** may form a portion of the external appearance of the refrigerator **1**. While the door **30** is in a closed position, the outer surface **31** of the door **30** may form a front surface of the door **30**.
(51) An inner surface **32** of the door **30** may be formed at a side opposite to the outer surface **31** of the door **30**. While the door **30** is in the closed position, the inner surface **32** of the door **30** may form a rear surface of the door **30**. While the door **30** is in the closed position, the inner surface **32** of the door **30** may be provided to face the inside of the main body **10**. While the door **30** is in the closed position, the inner surface **32** of the door **30** may be provided to cover the front of the storage compartment **20**.
(52) The door **30** may include a door insulator **35**. The door insulator **35** may be provided between the outer surface **31** of the door **30** and the inner surface **32** of the door **30**. For example, a foaming

space may be formed between the outer surface **31** of the door **30** and the inner surface **32** of the door **30**, and the door insulator **35** may be foamed in the foaming space. The door insulator **35** may prevent a heat exchange from occurring between the outer surface **31** of the door **30** and the inner surface **32** of the door **30**. The door insulator **35** may improve insulation performance between the inside of the storage compartment **20** and the outside of the door **30**.

(53) Urethane foam insulation, expanded polystyrene insulation (EPS), a vacuum insulation panel, and the like may be used for the door insulator **35**. However, the disclosure is not limited thereto, and the door insulator **35** may include various other materials to improve insulation performance between the storage compartment **20** and the door **30**.

(54) For example, the door insulator **35** may be configured with an insulator formed of the same material as a main body insulator **55** to be descried below. On the other hand, for example, the door insulator **35** may be configured with an insulator formed of a different material from the main body insulator **55**.

(55) The door **30** may include a door gasket **33**. The door gasket **33** may be provided on the inner surface **32** of the door **30**. The door gasket **33** may seal a gap between the door **30** and the main body **10** to prevent leakage of cold air from the storage compartment **20**. The door gasket **33** may be provided along the periphery of the inner surface **32** of the door **30**. The door gasket **33** may be disposed to be parallel to the opening **10***a* of the main body **10** while the door **30** is closed. The door gasket **33** may be configured to include an elastic material such as rubber. However, it is not limited thereto, and the door gasket **33** may include various materials to seal between the door **30** and the main body **10**.

(56) The door **30** may include a door basket **34** for storing food. The door basket **34** may be provided on the inner surface **32** of the door **30**.

(57) The door **30** may be provided to open and close the inner space of the main body **10** as a whole. The door **30** may be provided to open and close the storage compartment **20** provided inside the main body **10**. The door **30** may be provided to open and close a first storage compartment **21** and a second storage compartment **22**, which will be described below. That is the first storage compartment **21** and the second storage compartment **22** may be opened and closed by a single door **30** without a need to provide a door for opening and closing the first storage compartment **21** and a door for opening and closing the second storage compartment **22**.

(58) For example, the door **30** may close the first storage compartment **21** to be described below, while the opening **10***a* of the main body **10** is closed. The door **30** may cover the front of the first storage compartment **21** while the opening **10***a* of the main body **10** is closed.

(59) For example, the door **30** may close the second storage compartment **22** to be described below while the opening **10***a* of the main body **10** is closed. The door **30** may cover the front of the second storage compartment **22** while the opening **10***a* of the main body **10** is closed.

(60) For example, the door **30** may cover the front of a first drawer **25** to be described below while the opening **10***a* of the main body **10** is closed. In a state in which the first drawer **25** is inserted into a first lower storage compartment **21***b*, the front of the first drawer **25** may be covered by the door **30**. However, it is not limited thereto, and the door **30** may directly close the first lower storage compartment **21***b*.

(61) For example, the door **30** may cover the front of a second drawer **26** to be described below while the opening **10***a* of the main body **10** is closed. In a state in which the second drawer **26** is inserted into the second storage compartment **22**, the front of the second drawer **26** may be covered by the door **30**. However, it is not limited thereto, and the door **30** may directly close the second storage compartment **22**.

(62) For example, the door **30** may cover the front of a first shelf **23** while the opening **10***a* of the main body **10** is closed. The entirety of the first shelf **23** may be disposed inside the main body **10**, and thus in a state in which the opening **10***a* of the main body **10** is closed, the first shelf **23** may be covered by the door **30** without being visible from the outside.

(63) For example, the door **30** may cover the front of a second shelf **24** while the opening **10**a of the main body **10** is closed. The entirety of the second shelf **24** may be disposed inside the main body **10**, and thus in a state in which the opening **10**a of the main body **10** is closed, the second shelf **24** may be covered by the door **30** without being visible from the outside.
(64) The main body **10** may include an outer case **50**. The outer case **50** may form at least a part of the external appearance of the refrigerator **1**. The outer case **50** may be provided on the outside of the inner case **60**. The outer case **50** may be formed to have a substantially box-like shape having an open front surface. For example, the outer case **50** may form upper and, lower surfaces, left, and right surfaces, and a rear surface of the refrigerator **1**.
(65) The outer case **50** may include a metal material. For example, the outer case **50** may be manufactured by processing a steel plate material.
(66) The main body **10** may include the inner case **60**. The inner case **60** may be provided on the inside of the outer case **50**. The inner case **60** may form the storage compartment **20**. The inner space of the inner case **60** may be provided as a storage compartment **20**. The inner case **60** may have a shape with an open front. The inner case **60** may be formed to have a substantially box-like shape having an open front surface.
(67) The inner case **60** may include inner walls **61**, **62**, **63**, **64**, and **65**.
(68) For example, the inner case **60** may include a right wall **61**, an upper wall **62**, a left wall **63**, a bottom wall **64**, and a rear wall **65**. For example, the right wall **61**, the upper wall **62**, the left wall **63**, the bottom wall **64**, and the rear wall **65** of the inner case **60** may form the storage compartment **20**.
(69) The inner case **60** may include a plastic material. For example, the inner case **60** may be manufactured by a vacuum forming process. For example, the inner case **60** may be manufactured by an injection molding process.
(70) The main body **10** may include the main body insulator **55**. The main body insulator **55** may be provided between the outer case **10** and the inner case **60**. The main body insulator **55** may be provided to insulate the outer case **50** and the inner case **60** from each other. As the main body insulator **55** is foamed between the outer case **50** and the inner case **60**, the outer case **50** and the inner case **60** may be coupled to each other. The main body insulator **55** may prevent heat exchange between the inside of the storage compartment **20** and the outside of the main body **10**, thereby improving the cooling efficiency of the storage compartment **20**.
(71) Urethane foam insulation, expanded polystyrene insulation (EPS), a vacuum insulation panel, and the like may be used for the main body insulator **55**. However, the disclosure is not limited thereto, and the main body insulator **55** may be configured to include various other materials.
(72) The main body **10** may include the storage compartment **20**. The storage compartment **20** may be formed by the inner case **60**. The storage compartment **20** may be provided as one space inside the inner case **60**. The storage compartment **20** may be provided as a single unit. For example, the inner case **60** may have a shape of a box with an open front, and include the storage compartment **20** formed therein.
(73) The storage compartment **20** may be divided into a plurality of spaces by a partition **70**. For example, the storage compartment **20** may be divided into the first storage compartment **21** and the second storage compartment **22** by the partition **70**.
(74) The main body **10** may include the first storage compartment **21** and the second storage compartment **22**. The first storage compartment **21** may be provided above the second storage compartment **22**. The first storage compartment **21** may be disposed upward (+Z direction) of the second storage compartment **22**. The second storage compartment **22** may be provided below the first storage compartment **21**. The second storage compartment **22** may be disposed downward (−Z direction) of the first storage compartment **21**.
(75) The first storage compartment **21** and the second storage compartment **22** may have different temperatures. However, it is not limited thereto, and the first storage compartment **21** and the

second storage compartment **22** may be provided to have the same temperature as needed.
(76) For example, the first storage compartment **21** may include a first upper storage compartment **21***a* and the first lower storage compartment **21***b*. The first upper storage compartment **21***a* may be provided above the first lower storage compartment **21***b*. The first upper storage compartment **21***a* may be disposed upward (+Z direction) of the first lower storage compartment **21***b*. The first lower storage compartment **21***b* may be provided below the first upper storage compartment **21***a*. The first lower storage compartment **21***b* may be disposed downward (−Z direction) of the first upper storage compartment **21***a*.
(77) For example, the first shelf **23** on which food may be placed, a storage container (not shown) in which food may be stored, and the like may be provided in the first upper storage compartment **21***a*. The first shelf **23** may be provided in at least one unit thereof. The storage container may be provided in at least one unit thereof.
(78) For example, the first drawer **25** may be provided in the first lower storage compartment **21***b*. The first drawer **25** may be provided to be withdrawn from and inserted into the first lower storage compartment **21***b* through the opening **10***a* of the main body **10**. The first drawer **25** may be provided to be withdrawn and inserted in the front-rear direction (X direction). For example, the second shelf **24** may be provided upward (+Z direction) of the first drawer **25**. Food may be placed on the second shelf **24**. For example, the first lower storage compartment **21***b* may be defined by the first drawer **25** and the second shelf **24**.
(79) Unlike the first upper storage compartment **21***a*, the first lower storage compartment **21***b* may be prevented from contacting external air even when the door **30** is opened with respect to the main body **10**. Thus, the first lower storage compartment **21***b* may be a space for food that needs to be stored for a relatively long time compared to the first upper storage compartment **21***a*.
(80) For example, the second drawer **26** may be provided in the second storage compartment **22**. The second drawer **26** may be provided to be withdrawn from or inserted into the second storage compartment **22** through the opening **10***a* of the main body **10**. The second drawer **26** may be provided to be withdrawn and inserted in the front and rear direction (X direction). For example, the partition **70** may be provided upward (+Z direction) of the second drawer **26**. For example, the second storage compartment **22** may be defined by the second drawer **26** and the partition **70**. However, the size of the second storage compartment **22** may vary depending on the location of the partition **70**.
(81) Meanwhile, the first storage compartment **21** may be provided as a refrigerating compartment **21**, and the second storage compartment **22** may be provided as a freezing compartment **22**. In this case, an ice maker **27** may be disposed inside the second storage compartment **22**. However, this is merely exemplary, and the first storage compartment **21** may be provided as a freezing compartment, and the second storage compartment **22** may be provided as a refrigerating compartment. In this case, the ice maker **27** may be disposed inside the first storage compartment **21**.
(82) Hereinafter, for convenience of description, a case in which the first storage compartment **21** is provided as a refrigerating compartment **21** and the second storage compartment **22** is provided as a freezing compartment **22** will be described as an example.
(83) The refrigerating compartment **21** may be provided to keep food refrigerated. The refrigerating compartment **21** may be maintained at an appropriate temperature for refrigerated storage.
(84) The freezing compartment **22** may be provided to keep food frozen. The freezing compartment **22** may be maintained at an appropriate temperature for frozen storage.
(85) The refrigerator **1** may include the partition **70**. The partition **70** may be disposed in the storage compartment **20**. The partition **70** may divide the storage compartment **20** into a plurality of spaces. The partition **70** may divide the storage compartment **20** into the refrigerating compartment **21** and the freezing compartment **22**.
(86) The partition **70** may be disposed in front of a cold air duct **80**. The partition **70** may be

detachably mounted on the cold air duct **80**. For example, the partition **70** may be detachably coupled to the front surface of the cold air duct **80**.
(87) The refrigerator **1** may include a cooling system provided to generate cold air using a cooling cycle and supply the generated cold air to the storage compartment **20**. The cooling system may generate cold air using latent heat of evaporation of the refrigerant in a cooling cycle.
(88) For example, the refrigerator **1** may include a compressor **11**. The refrigerator **1** may include a condenser (not shown). The refrigerator **1** may include an expansion valve (not shown). The refrigerator **1** may include an evaporator **12**. The refrigerator **1** may include a fan **13**.
(89) A cooling room **20***b* may be provided in the main body **10**. The evaporator **12**, the fan **13**, and the like may be provided in the cooling room **20***b*. The cooling room **20***b* may be referred to as a cold air forming space **20***b*.
(90) A machine room **15** may be provided in the main body **10**. The compressor **11** and the condenser may be provided in the machine room **15**.
(91) Parts of the refrigerator **1** constituting the cooling system may have a relatively great weight. Accordingly, the cooling room **20***b* and the machine room **15** may be provided in a lower portion of the main body **10**. However, it is not limited thereto, and the cooling room **20***b* and the machine room **15** may be arranged in various forms, and the components constituting the cooling system may be arranged in various forms to correspond to the positions of the cooling room **20***b* and the machine room **15**.
(92) Since cold air is generated by the evaporator **12** in the cooling room **20***b*, the cooling room **20***b* may be maintained at a relatively low temperature. Alternatively, since heat is generated by the compressor **11** and the condenser in the machine room **15**, the machine room **15** may be maintained at a relatively high temperature. Accordingly, the cooling room **20***b* and the machine room **15** may be formed in respective spaces separated from each other to be thermally insulated. For example, the main body insulator **55** may be foamed between the cooling room **20***b* and the machine room **15**.
(93) The evaporator **12** provided in the cooling room **20***b* may generate cold air by evaporating a refrigerant, and the cold air generated by the evaporator **12** may be caused to flow through the fan **13**. Cold air caused to flow by the fan **13** may be supplied to the storage compartment **20**. The evaporator **12** may generate cold air in the cooling room **20***b*, and the cold air generated by the evaporator **12** may be caused to flow from the cooling room **20***b* to the storage compartment **20** by the fan **13**.
(94) The refrigerator **1** may be an indirect cooling refrigerator. For convenience of description, the refrigerator **1** according to an embodiment of the disclosure is described on the assumption that the refrigerator **1** is an indirect cooling refrigerator, but the concept of the disclosure is not limited thereto and may be applied to a direct cooling refrigerator.
(95) The evaporator **12**, the fan **13**, and the like disposed in the cooling room **20***b* may be referred to as a cold air supply device as generating cold air and causing the cold air to flow such that the storage compartment **20** is supplied with the cold air.
(96) The refrigerator **1** may include a damper **14**. The damper **14** may be accommodated inside the cold air duct **80**. The damper **14** may be provided to open and close a passage formed inside the cold air duct **80**. The damper **14** may be provided to adjust the opening degree of the passage formed inside the cold air duct **80**.
(97) The refrigerator **1** may include the cold air duct **80**. The cold air duct **80** may form a passage through which cold air may flow. The cold air duct **80** may be provided to collect cold air inside the storage compartment **20** and guide the cold air to the evaporator **12**. The cold air duct **80** may be provided to guide cold air generated by the evaporator **12** into the storage compartment **20**. The cold air duct **80** may be provided to supply cold air generated by the evaporator **12** to the refrigerating compartment **21** and the freezing compartment **22**.
(98) The refrigerator **1** may include the plate assembly **80***a*. For example, the cold air duct **80** of the

refrigerator **1** may include the plate assembly **80***a*. However, this is merely exemplary, and the plate assembly **80***a* may be applied for coupling various parts. For example, the main body **10** of the refrigerator **1** may include the plate assembly **80***a*. For example, the door **30** of the refrigerator **1** may include the plate assembly **80***a*. In addition, as described above, the plate assembly **80***a* may be applied to various home appliances other than the refrigerator **1**.
(99) For example, the plate assembly **80***a* may form at least a part of the cold air duct **80**. For example, the plate assembly **80***a* may form a passage that allows cold air to flow therein.
(100) FIG. **3** is a perspective view illustrating an example of a plate assembly according to one embodiment. FIG. **4** is a perspective view illustrating the plate assembly shown in FIG. **3** when viewed from the rear. FIG. **5** is an exploded perspective view illustrating the plate assembly shown in FIG. **3**. FIG. **6** is an exploded perspective view illustrating the plate assembly shown in FIG. **5** from the rear.
(101) The plate assembly **80***a* may include a first plate **100**. The plate assembly **80***a* may include a second plate **200**.
(102) The first plate **100** and the second plate **200** may be provided to face each other. The first plate **100** and the second plate **200** may be provided opposed to each other in the front-back direction (X direction).
(103) The first plate **100** may be coupled to the front of the second plate **200**. The first plate **100** may be detachably coupled to the front of the second plate **200**.
(104) For example, the first plate **100** may be exposed to the outside of the home appliance. For example, the first plate **100** may be exposed toward the storage compartment **20**. For example, the first plate **100** may be provided to form a front external appearance of the plate assembly **80***a*. The first plate **100** may be referred to as a front plate **100**.
(105) The first plate **100** may include a first body **110**. The first body **110** may form the external appearance of the first plate **100**. The first body **110** may be referred to as a front body **110**.
(106) The first plate **100** may include a first coupling portion **120** for coupling with the second plate **200**. The first coupling portion **120** may be formed on a side of the first body **110** facing the second plate **200**. The first coupling portion **120** may be formed on a surface of the first body **110** facing the second plate **200**. For example, when a support member **300** to be described below is disposed between the first plate **100** and the second plate **200**, the first coupling portion **120** may be formed on one surface facing the support member **300** of the first body **110**. For example, the one surface of the first body **110** facing the support member **300** may be an inner surface of the first body **110**. For example, a surface provided on a side opposite to the inner surface of the first body **110** may be an outer surface. The first coupling portion **120** of the first plate **100** may be provided to correspond to a second coupling portion **220** of the second plate **200** to be described below. The first coupling portion **120** may be provided to be coupled to the second coupling portion **220**. The first coupling portion **120** may be detachably coupled to the second coupling portion **220**.
(107) The first coupling portion **120** may include a protruding portion protruding from the first body **110** in the first direction A. The protruding portion may be inserted into an opening **221** of the second coupling portion **220** to be described below. The protruding portion may be locked with a locking portion **222** of the second coupling portion **220** to be described below.
(108) The protruding portion of the first coupling portion **120** may include a first coupling protrusion **121** extending in a first direction A toward the second plate **200** from the first body **110**. The first coupling protrusion **121** may be inserted into the opening **221** of the second coupling portion **220**. The first coupling protrusion **121** may be locked to the locking portion **222** of the second coupling portion **220**.
(109) For example, the first coupling protrusion **121** may include a first ridge **121***a* configured to be locked with the locking portion **222** of the second coupling portion **220**. For example, the first ridge **121***a* may be formed at the end of the first coupling protrusion **121**.
(110) The protruding portion of the first coupling portion **120** may include a second coupling

protrusion **122** spaced apart from the first coupling protrusion **121**. The second coupling protrusion **122** may extend in the first direction A toward the second plate **200** from the first body **110**. The second coupling protrusion **122** may be inserted into the opening **221** of the second coupling portion **220**. The second coupling protrusion **122** may be locked with the locking portion **222** of the second coupling portion **220**.
(111) For example, the second coupling protrusion **122** may include a second ridge **122*a*** configured to be locked with by the locking portion **222** of the second coupling portion **220**. For example, the second ridge **122*a*** may be formed at the end of the second coupling protrusion **122**.
(112) For example, the first coupling protrusion **121** and the second coupling protrusion **122** may be provided as a pair.
(113) For example, the first plate **100** may include a base portion **129** formed on a side of the first body **110** facing the second plate **200**. For example, the first coupling protrusion **121** and the second coupling protrusion **122** may protrude from the base portion **129**.
(114) The first plate **100** may include at least one groove **127** or **128**. A worker's hand or tool T may be inserted into the at least one groove **127** or **128**.
(115) For example, the first plate **100** may include a first groove **127** formed on the first coupling protrusion **121**. For example, the first groove **127** may be formed at the end of the first coupling protrusion **121**.
(116) For example, the first plate **100** may include a second groove **128** formed on the second coupling protrusion **122**. For example, the second groove **128** may be formed at the end of the second coupling protrusion **122**.
(117) The first coupling portion **120** may include a bridge **123**. The bridge **123** may connect the first coupling protrusion **121** and the second coupling protrusion **122**. The bridge **123** may include a recessed portion **124** recessed in a second direction B opposite to the first direction A. For example, the recessed portion **124** may be formed at the center of the bridge **123**. For example, the bridge **123** may have a substantially V shape.
(118) For example, the bridge **123** may include a first bridge portion **125** extending from the first coupling protrusion **121** to the recessed portion **124**. The first bridge portion **125** may have a downwardly sloping shape toward the recessed portion **124**.
(119) For example, the bridge **123** may include a second bridge portion **126** extending from the second coupling protrusion **121** to the recessed portion **124**. The second bridge portion **126** may have a downwardly sloping shape toward the recessed portion **124**.
(120) The first coupling portion **120** of the first plate **100** may be provided to be elastically deformable. For example, the protruding portion may be provided to be elastically deformable. For example, the first coupling protrusion **121** may be provided to be elastically deformable. For example, the second coupling protrusion **122** may be provided to be elastically deformable. For example, the bridge **123** may be provided to be elastically deformable.
(121) The second plate **200** may be coupled to the rear of the first plate **100**. The second plate **200** may be detachably coupled to the rear of the first plate **100**.
(122) For example, the second plate **200** may be covered by the first plate **100** and thus not exposed to the outside of the home appliance. For example, the second plate **200** may be provided to face the rear surface of the inner case **60**. For example, the second plate **200** may be disposed to face the rear wall **65** of the inner case **60**. For example, the second plate **200** may form the rear external appearance of the plate assembly **80*a***. The second plate **200** may be referred to as a rear plate **200**.
(123) The second plate **200** may include a second body **210**. The second body **210** may form the external appearance of the second plate **200**. The second body **210** may be referred to as a rear body **210**.
(124) The second plate **200** may include the second coupling portion **220** for coupling with the first plate **100**. The second coupling portion **220** may be formed on a side of the second body **210** facing the first plate **100**. The second coupling portion **220** of the second plate **200** may be provided to

correspond to the first coupling portion **120** of the first plate **100**. The second coupling portion **220** may be detachably coupled to the first coupling portion **120**. The second coupling portion **220** may be provided to be locked with the first coupling portion **120**.

(125) The second coupling portion **220** may include the opening **221**. The opening **221** may be formed in the second body **210** such that at least a portion of the first coupling portion **120** is inserted. The opening **221** may be provided to allow the protruding portion to be inserted thereinto. The opening **221** may be provided to allow the first coupling protrusion **121** to be inserted thereinto. The opening **221** may be provided to allow the second coupling protrusion **122** to be inserted thereinto.

(126) The second coupling portion **220** may include the locking portion **222**. The locking portion **222** may be provided to be coupled to the first coupling portion **120** inserted into the opening **221**. The locking portion **222** may be provided to be locked with the first coupling portion **120** inserted into the opening **221**. For example, the locking portion **222** may protrude radially inward of the opening **221**. The first ridge **121***a* of the first coupling protrusion **121** may be provided to be locked with the locking portion **222**. The second ridge **122***a* of the second coupling protrusion **122** may be provided to be locked with the locking portion **222**.

(127) The second coupling portion **220** may include a coupling body **223**. The coupling body **223** may be provided to surround at least a portion of the first coupling portion **120**. The coupling body **223** may be provided to accommodate at least a portion of the first coupling portion **120**. For example, the coupling body **223** may be provided to surround an end of the first coupling protrusion **121**, an end of the second coupling protrusion **122**, and at least a portion of the bridge **123**. For example, the coupling body **223** may extend along the direction in which the first plate **100** and the second plate **200** are arranged. For example, the coupling body **223** may have a substantially sidewall shape. For example, the coupling body **223** may be provided to form the opening **221**. For example, the locking portion **222** may protrude toward the opening **221** from the coupling body **223**.

(128) As the first coupling portion **120** and the second coupling portion **220** are coupled to each other, the first plate **100** and the second plate **200** may be assembled with each other. The first plate **100** may include one or more first coupling portions **120**, and the second plate **200** may include one or more second coupling portions **220**. For example, the first coupling portions **120** may be provided corresponding in number to the number of the second coupling portions **220**.

(129) The second plate **200** may include a slit **230**. The slit **230** may be formed in the second body **210**. The slit **230** may be provided to pass through the second body **210**. The slit **230** may be disposed adjacent to the second coupling portion **220**. The slit **230** may be spaced apart from the opening **221**. The slit **230** may be disposed radially outward of the opening **221**.

(130) The slit **230** may allow a force to act on the second plate **200** in a direction to be separated from the first plate **100** in a state in which the first plate **100** and the second plate **200** are coupled. The slit **230** may allow a force to act on the second body **210** in a direction away from the first body **110** in a state in which the first plate **100** and the second plate **200** are coupled to each other. Accordingly, the plate assembly **80***a* may be provided to have a repulsive force in a direction to separate the first plate **100** and the second plate **200** from each other in a state in which the first plate **100** and the second plate **200** are assembled with each other. As a result, the first plate **100** and the second plate **200** of the plate assembly **80***a* may be easily disassembled through a repulsive force. A worker may disassemble the first plate **100** and the second plate **200** without exerting a great force. Details thereof will be described below.

(131) The second plate **200** may be provided to be elastically deformable. For example, as the second plate **200** includes the slit **230**, at least a portion of the second body **210** may be provided to be elastically deformable. For example, a portion of the second body **210** adjacent to the slit **230** may be provided to be elastically deformable. For example, the second coupling portion **220** of the second plate **200** may be provided to be elastically deformable.

(132) Meanwhile, the first plate **100** and the second plate **200** are not limited in the configuration by the ordinal numbers of "first" and "second". For example, the first plate **100** could be referred to as the second plate, while the second plate **200** could be referred to as the first plate.
(133) The plate assembly **80***a* may further include the support member **300**. The support member **300** may be provided between the first plate **100** and the second plate **200**. The support member **300** may be provided to support at least one of the first plate **100** or the second plate **200**. For example, the support member **300** may be provided to be in surface contact with at least one of the first plate **100** or the second plate **200**. The support member **300** may be referred to as an intermediate member **300**.
(134) For example, the support member **300** may include at least one of an expanded polystyrene insulation, a urethane foam insulation, or a vacuum insulation panel. The support member **300** may include various heat insulations for improving the heat insulating performance of the cold air duct **80**. In this case, the support member **300** may be referred to as a heat insulating member **300**. However, it is not limited thereto, and depending on the characteristics of the home appliance to which the plate assembly **80***a* is applied, the support member **300** may include various materials.
(135) The support member **300** may include a third body **310**. The third body **310** may form the external appearance of the support member **300**. A first surface **311** of the third body **310** may be provided to face the first body **110** of the first plate **100**. The first surface **311** of the third body **310** may be provided to be in surface contact with the first body **110**. A second surface **312** of the third body **310** is aside opposite to the first surface **311** and may be provided to face the second body **210** of the second plate **200**. The second surface **312** of the third body **310** may be provided to be in surface contact with the second body **210**.
(136) The third body **310** may be referred to as a support body **310**. The third body **310** may be referred to as an intermediate body **310**.
(137) The support member **300** may include a support hole **320**. The support hole **310** may correspond to the opening **221** of the second coupling portion **220**. The support hole **310** may be provided to accommodate the first coupling portion **120**. The support hole **310** may be provided to surround the first coupling portion **120**.
(138) The support member **300** may include a seating portion **330** extending from the support hole **320**. The seating portion **330** may be provided to support at least a portion of the second coupling portion **220**. For example, the seating portion **330** may be provided to support the locking portion **222** of the second coupling portion **220**. For example, the seating portion **330** may be provided to form a step with respect to the support hole **320**. For example, the seating portion **330** may be provided to form a step with respect to the second surface **312** of the third body **310**.
(139) FIG. **7** is an enlarged view of a portion illustrating an example of a plate assembly according to an embodiment. FIG. **8** is a cut-away perspective view illustrating a state before a plate assembly according to one embodiment is assembled. FIG. **9** is a cutaway perspective view illustrating a state in which the plate assembly shown in FIG. **8** is assembled.
(140) Referring to FIG. **7**, in a state in which the first plate **100** and the second plate **200** are coupled to each other, at least a portion of the first coupling portion **120** may be provided to be exposed through the opening **221**. For example, the end of the first coupling protrusion **121** may be exposed through the opening **221**. For example, the end of the second coupling protrusion **122** may be exposed through the opening **221**.
(141) In a state in which the first plate **100** and the second plate **200** are coupled to each other, at least a portion of the first coupling portion **120** may be provided to be accessible through the opening **221**. For example, the first groove **127** may be provided to be accessible through the opening **221**. For example, the second groove **128** may be provided to be accessible through the opening **221**. For example, a worker's hand or tool may be inserted into at least one of the first groove **127** or the second groove **128** through the opening **221**.
(142) The slit **230** of the second plate **200** may extend along a third direction C. The third direction

C may be a direction in which the first coupling protrusion **121** and the second coupling protrusion **122** are arranged.

(143) The slit **230** may include a first slit portion **231** and a second slit portion **232** spaced apart from the first slit portion **231**. For example, in a state in which the first plate **100** and the second plate **200** are coupled to each other, the first coupling portion **120** is provided between the first slit portion **231** and the second slit portion **232**.

(144) For example, the first slit portion **231** and the second slit portion **232** may be provided to be symmetrical with respect to a first line L**1**. The first line L**1** may pass through the center of the opening **221** while extending along the third direction C. For example, the first slit portion **231** and the second slit portion **232** may be provided as a pair.

(145) For example, the slit **230** may include a first slit forming portion **230***a* extending along a portion of an edge of the coupling body **223**. For example, the first slit forming portion **230***a* may have a shape corresponding to a portion of an edge of the coupling body **223**. For example, the first slit forming portion **230***a* may have a curved shape. However, it is not limited thereto, and when the edge of the coupling body **223** is provided in a straight line, the first slit forming portion **230***a* may include a straight line shape.

(146) For example, the slit **230** may include a second slit forming portion **230***b* extending from one side of the first slit forming portion **230***a* along the third direction C. For example, the second slit forming portion **230***b* may extend linearly along the third direction C. For example, the second slit forming portion **230***b* may have a straight line shape.

(147) For example, the slit **230** may include a third slit forming portion **230***c* extending from the other side of the first slit forming portion **230***a* along the third direction C. For example, the third slit forming portion **230***c* may extend linearly along the third direction C. For example, the third slit forming portion **230***b* may have a straight line shape.

(148) For example, each of the first slit portion **231** and the second slit portion **232** may include a first slit forming portion **230***a*, a second slit forming portion **230***b*, and a third slit forming portion **230***c*.

(149) For example, the edge of the coupling body **223** of the second coupling portion **220** may include a first portion **2231** adjacent to the slit **230** and a second portion **2232** other than the first portion **2231**. The first portion **2231** may be provided adjacent to the first slit forming portion **230***a*. The second portion **2232** may be provided to be connected to the second body **210**.

(150) Referring to FIG. **8**, for coupling of the first plate **100** and the second plate **200**, the first plate **100** and the second plate **200** may be aligned. The support member **300** may be provided between the first plate **100** and the second plate **200**. However, it is not limited thereto, and the support member **300** may be omitted.

(151) The first coupling portion **120** of the first plate **100** may protrude toward the second coupling portion **220** of the second plate **200**. The first body **110** of the first plate **100** may be provided to face the first surface **311** of the third body **310** of the support member **300**. The first coupling portion **120** of the first plate **100** may be inserted into the support hole **320** of the support member **300**.

(152) The second coupling portion **220** of the second plate **200** may correspond to the first coupling portion **120** of the first plate **100**. The second coupling portion **220** of the second plate **200** may correspond to the support hole **320** and the seating portion **330** of the support member **300**. The second body **210** of the second plate **200** may be provided to face the second surface **312** of the third body **310** of the support member **300**.

(153) Referring to FIG. **9**, the first coupling portion **120** and the second coupling portion **220** may be coupled to each other. The first coupling portion **120** may be inserted into the support hole **320** and coupled to the second coupling portion **220**. The first coupling portion **120** may be locked with the second coupling portion **220**. As the first coupling portion **120** and the second coupling portion **220** are coupled to each other, the first plate **100** and the second plate **200** may be assembled with

each other.
(154) The first coupling protrusion **121** may be inserted into the opening **221** and coupled to the locking portion **222**. The first coupling protrusion **121** may be inserted into the opening **221** and locked with the locking portion **222**. The first ridge **121***a* of the first coupling protrusion **121** may be provided to be locked with the locking portion **222**. The first coupling protrusion **121** may be restricted from moving in the first direction A and/or the second direction B while being locked with the locking portion **222**.
(155) The second coupling protrusion **122** may be inserted into the opening **221** and coupled to the locking portion **222**. The second coupling protrusion **122** may be inserted into the opening **221** and locked with the locking portion **222**. The second ridge **122***a* of the second coupling protrusion **122** may be provided to be locked with the locking portion **222**. The second coupling protrusion **122** may be restricted from moving in the first direction A and/or the second direction B while being locked with the locking portion **222**.
(156) In general, for example, a screw fastening method may be used for assembly between parts. In this case, separate equipment (e.g., a screwdriver, etc.) may be required for screw fastening, and the work time may be relatively long. When the screw fastening is excessive, the exterior of the parts may be depressed. For example, when the thickness of the intermediate member is thick, the exterior of the parts may be damaged during screw fastening.
(157) As another example, a hook method may be used for assembling parts. In this case, when one hook is assembled, another hook that has already been assembled may be disassembled. That is, inconvenience of having to reassemble the hook may be caused. In addition, when the hook fastening is excessive, the exterior of the parts may be depressed. For example, when the thickness of the intermediate member is thick, the exterior of the parts may be damaged during the hook assembly process.
(158) According to the disclosure, the first plate **100** and the second plate **200** may be easily coupled to each other. The first coupling portion **120** and the second coupling portion **220** may be coupled to each other without a separate fastening member (e.g., a screw, a screwdriver, etc.). The first coupling portion **120** and the second coupling portion **220** may be provided to enable one-touch coupling. For example, the operator may assemble the first coupling portion **120** and the second coupling portion **220** by pressing one of the first coupling portion **120** and the second coupling portion **220** using a finger or an assembly jig. In addition, in a state in which the first coupling portion **120** and the second coupling portion **220** are locked with each other, the coupling of the first coupling portion **120** and the second coupling portion may be firmly maintained without a predetermined force applied to the first coupling portion **120** and the second coupling portion **220**. As described above, the coupling of the first plate **100** and the second plate **200** may be facilitated, and the working time may be shortened. As a result, assemblability and productivity of the plate assembly **80***a* may be improved.
(159) Also, the first plate **100** may be exposed to the outside of the home appliance, and the second plate **200** may be covered by the first plate **100** without being exposed to the outside. Since the second plate **200** includes the slit **230**, the rigidity of the second plate **200** may be weaker than that of the first plate **100**. For example, in a state in which the first plate **100** and the second plate **200** are coupled to each other, stress may be concentrated on an end of the slit **230** and/or a portion of the coupling body **223**. For example, stress may be concentrated on the end of the second slit forming portion **230***b*, the end of the third slit forming portion **230***c*, and the second portion **2232** of the coupling body **223**. That is, stress may not be concentrated on the first plate **100**. Accordingly, when the first coupling portion **120** and the second coupling portion **220** are coupled to each other, only the second plate **200** having relatively weak rigidity may be deformed. As a result, an external appearance of the first plate **100** exposed to the outside of the home appliance may not be depressed. Aesthetics of the plate assembly **80***a* may not be degraded.
(160) The support member **300** may be provided to be in surface contact with at least one of the

first plate **100** or the second plate **200**. As the first coupling portion **120** and the second coupling portion **220** are coupled to each other, the first plate **100** and the support member **300** may come into close contact with each other. As the first coupling portion **120** and the second coupling portion **220** are coupled to each other, the second plate **200** and the support member **300** may come into close contact with each other. For example, even when the thickness of the support member **300** is small, the second coupling portion **220** may be coupled to the first coupling portion **120** while being deformed. Accordingly, the airtightness of the plate assembly **80***a* may be improved. For example, when the plate assembly **80***a* is applied to the cold air duct **80**, cold air flowing inside the cold air duct **80** may not flow outside, or wet air may not flow into the cold air duct **80**.
(161) FIG. **10** is a cross-sectional view illustrating a state in which a first plate and a second plate of a plate assembly are coupled according to an embodiment. FIG. **11** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **10** are being disassembled with each other. FIG. **12** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **11** are disassembled with each other.
(162) An example of a process of disassembling the plate assembly **80***a* is be described with reference to FIGS. **10** to **12**.
(163) Referring to FIG. **10**, in a state in which the first coupling portion **120** and the second coupling portion **220** are coupled to each other, at least a portion of the first coupling portion **120** may be provided to be accessible through the opening **221** of the second coupling portion **220**. For example, a tool T**1** may be inserted into the groove **127** of the first coupling protrusion **121** through the opening **221**. For example, the tool T**1** may be inserted into the groove **128** of the second coupling protrusion **122** through the opening **221**. For example, the tool T**1** may have a substantially tong shape. For example, the tool T**1** may be a long nose plier. Also, although not shown in the drawings, the operator may insert a finger into at least one of the grooves **127** or **128** instead of the tool T**1**.
(164) Referring to FIG. **11**, in a state in which the first plate **100** and the second plate **200** are coupled to each other, the first coupling protrusion **121** and the second coupling protrusion **122** may be pressed to be brought close together. The first coupling protrusion **121** and the second coupling protrusion **122** may be provided to be closed by the tool T**1**, a worker's finger, or the like. As the first coupling protrusion **121** and the second coupling protrusion **122** are brought close together, the bridge **123** may be elastically deformed. The first bridge portion **125** and the second bridge portion **126** may be brought close together. Accordingly, the coupling between the first coupling portion **120** and the second coupling portion **220** may be released. The first coupling portion **120** and the second coupling portion **220** may be separated from each other. Locking of the first coupling portion **120** and the second coupling portion **220** may be released. The coupling between the first coupling protrusion **121** and the locking portion **222** may be released. The coupling between the second coupling protrusion **122** and the locking portion **222** may be released.
(165) Referring to FIG. **12**, as the coupling between the first coupling portion **120** and the second coupling portion **220** is released, the first plate **100** and the second plate **200** may be disassembled from with each other. As the locking of the first coupling portion **120** and the second coupling portion **220** is released, the second plate **200** may be provided to move away from the first plate **100** by a repulsive force. In a state in which the first plate **100** and the second plate **200** are coupled to each other, the slit **230** may cause a force to act on the second plate **200** in a direction to be separated from the first plate **100**. For example, as an external force presses on the first coupling portion **120** and the first coupling portion **120** and the second coupling portion **220** are decoupled, the second coupling portion **220** may be biased in a direction away from the first plate **100** by the slit **230**. Accordingly, when the first coupling portion **120** and the second coupling portion **220** are separated from each other, the slit **230** may be provided to allow the second plate **200** to bounce against, or spring away from, the first plate **100**. For example, in a state in which the first plate **100**

and the second plate **200** are coupled to each other, when the first coupling portion **120** is deformed along the third direction C, the second portion (**2232** in FIG. **7**) of the coupling body **223** may be provided to bounce in the first direction A. As a result, disassembly of the first coupling portion **120** and the second coupling portion **220** may be facilitated. The first coupling portion **120** and the second coupling portion **220** may be provided to enable one-touch disassembly.
(166) FIG. **13** is a cross-sectional view illustrating a state in which a first plate and a second plate of a plate assembly are coupled according to an embodiment. FIG. **14** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **13** are being disassembled from with each other. FIG. **15** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **14** are disassembled from with each other.
(167) An example of a process of disassembling the plate assembly **80***a* is described with reference to FIGS. **13** to **15**.
(168) Referring to FIG. **13**, in a state in which the first coupling portion **120** and the second coupling portion **220** are coupled to each other, at least a portion of the first coupling portion **120** may be provided to be accessible through the opening **221** of the second coupling portion **220**. For example, a tool T**2** may be inserted into the recessed portion **124** of the bridge **123** through the opening **221**. For example, the tool T**2** may have a shape extending in approximately one direction. For example, the tool T**2** may be a slotted screwdriver. In addition, although not shown in the drawings, the operator may insert a finger into the recessed portion **124** of the bridge **123** instead of the tool T**2**.
(169) Referring to FIG. **14**, in a state in which the first plate **100** and the second plate **200** are coupled to each other, the bridge **123** may be pressed. In a state in which the first plate **100** and the second plate **200** are coupled to each other, the recessed portion **124** of the bridge **123** may be pressed in the second direction B. As the recessed portion **124** of the bridge **123** is pressed in the second direction B, the first coupling protrusion **121** and the second coupling protrusion **122** may be brought close together. As the recessed portion **124** of the bridge **123** is pressed in the second direction B, the first coupling protrusion **121** and the second coupling protrusion **122** may be provided to be closed. The first coupling protrusion **121** and the second coupling protrusion **122** may be elastically deformed. Accordingly, the coupling between the first coupling portion **120** and the second coupling portion **220** may be released. The first coupling portion **120** and the second coupling portion **220** may be separated from each other. Locking of the first coupling portion **120** and the second coupling portion **220** may be released. The coupling between the first coupling protrusion **121** and the locking portion **222** may be released. The coupling between the second coupling protrusion **122** and the locking portion **222** may be released.
(170) Referring to FIG. **15**, as the coupling between the first coupling portion **120** and the second coupling portion **220** is released, the first plate **100** and the second plate **200** may be disassembled from with each other. As the locking of the first coupling portion **120** and the second coupling portion **220** is released, the second plate **200** may be provided to move away from the first plate **100** by a repulsive force. In a state in which the first plate **100** and the second plate **200** are coupled to each other, the slit **230** may cause a force to act on the second plate **200** in a direction to be separated from the first plate **100**. Accordingly, in response to the first coupling portion **120** and the second coupling portion **220** being separated from each other, the slit **230** may be provided to allow the second plate **200** to bounce against the first plate **100**. For example, in a state in which the first plate **100** and the second plate **200** are coupled to each other, when the first coupling portion **120** is deformed along the second direction B, the second portion (**2232** in FIG. **7**) of the coupling body **223** may be provided to bounce in the first direction A. As a result, disassembly of the first coupling portion **120** and the second coupling portion **220** may be facilitated. The first coupling portion **120** and the second coupling portion **220** may be provided to enable one-touch disassembly.
(171) FIG. **16** is an enlarged view illustrating a portion of an example of a plate assembly

according to one embodiment. FIG. **17** is a cross-sectional view of the plate assembly shown in FIG. **16**. FIG. **18** is a cross-sectional view illustrating a state in which the first plate and the second plate of the plate assembly shown in FIG. **17** are being disassembled from with each other.

(172) An example of a process of disassembling the plate assembly is described with reference to FIGS. **16** to **18**. The shape of the slit of the plate assembly shown in FIGS. **16** to **18** may be different from the shape of the slit of the plate assembly shown in FIGS. **5** to **15**. Other configurations may be the same as those of the plate assembly described with reference to FIGS. **5** to **15**. In describing the plate assembly shown in FIGS. **16** to **18**, the same reference numerals are assigned to the same configurations as those of the plate assembly **80***a* described with reference to FIGS. **5** to **15**, and detailed description thereof may be omitted.

(173) Referring to FIG. **16**, the slit **230** may include a first slit portion **231***a* and a second slit portion **231***b* spaced apart from the first slit portion **231***a*. For example, in a state in which the first plate **100** and the second plate **200** are coupled to each other, the first coupling portion **120** may be provided between the first slit portion **231***a* and the second slit portion **231***b*.

(174) For example, the first slit portion **231***a* and the second slit portion **231***b* may be provided to be symmetrical with respect to a second line L**2**. The second line L**2** may pass through the center of the opening **221** while extending along a fourth direction D crossing the third direction C. For example, the fourth direction D may be perpendicular to the third direction C. For example, the first slit portion **231***a* and the second slit portion **231***b* may be provided as a pair.

(175) For example, the slit **230** may include a fourth slit forming portion **230***d* spaced apart from the opening **221** and extending along the fourth direction D. For example, the fourth slit forming portion **230***d* may be spaced apart from the opening **221** by a predetermined distance along the third direction C. For example, the fourth slit forming portion **230***d* may linearly extend along the fourth direction D. For example, the fourth slit forming portion **230***d* may have a straight line shape.

(176) For example, the slit **230** may include a fifth slit forming portion **230***e* extending from one side of the fourth slit forming portion **230***d* toward the opening **221** along the third direction C. For example, the fifth slit forming portion **230***e* may extend linearly along the third direction C. For example, the fifth slit forming portion **230***e* may have a straight line shape.

(177) For example, the slit **230** may include a sixth slit forming portion **230***f* extending from the other side of the fourth slit forming portion **230***d* toward the opening **221** along the third direction C. For example, the sixth slit forming portion **230***f* may extend linearly along the third direction C. For example, the sixth slit forming portion **230***f* may have a straight line shape.

(178) For example, each of the first slit portion **231***a* and the second slit portion **231***b* may include a fourth slit forming portion **230***d*, a fifth slit forming portion **230***e*, and a sixth slit forming portion **230***f*.

(179) The second plate **200** may include a wing portion **240**. The slit **230** of the second plate **200** may form the wing portion **240**. The slit **230** of the second plate **200** may be provided to define the wing portion **240**. For example, the wing portion **240** may be a portion surrounded by the slit **230** and the coupling body **223**. For example, the wing portion **240** may be provided to be elastically deformable.

(180) For example, the second plate **200** may include a first wing portion **241** formed by the first slit portion **231***a*. For example, the first wing portion **241** may be formed by the fourth slit forming portion **230***d*, the fifth slit forming portion **230***e*, and the sixth slit forming portion **230***f* of the first slit portion **231***a*. The first wing portion **241** may have a cantilever shape.

(181) For example, the second plate **200** may include a second wing portion **242** formed by the second slit portion **231***b*. For example, the second wing portion **242** may be formed by the fourth slit forming portion **230***d*, the fifth slit forming portion **230***e*, and the sixth slit forming portion **230***f* of the second slit portion **231***b*. The second wing **242** may have a cantilever shape.

(182) For example, the first wing portion **241** and the second wing portion **242** may be provided to

be symmetrical with respect to the second line L**2**.
(183) Referring to FIG. **17**, the first coupling portion **120** and the second coupling portion **220** may be coupled to each other. The first coupling portion **120** may be inserted into the support hole **320** and coupled to the second coupling portion **220**. The first coupling portion **120** may be locked with the second coupling portion **220**. The first coupling protrusion **121** may be inserted into the opening **221** and locked with the locking portion **222**. The second coupling protrusion **122** may be inserted into the opening **221** and locked with the locking portion **222**. In a state in which the first coupling portion **120** and the second coupling portion **220** are locked with each other, disassembly of the first plate **100** and the second plate **200** may be prevented.
(184) Referring to FIG. **18**, as the locking of the first coupling portion **120** and the second coupling portion **220** is released, the first plate **100** and the second plate **200** may be disassembled from with each other. As the locking of the first coupling portion **120** and the second coupling portion **220** is released, the second plate **200** may be provided to move away from the first plate **100** by a repulsive force. For example, as the first coupling protrusion **121** and the second coupling protrusion **122** are brought close together r, locking of the first coupling portion **120** and the second coupling portion **220** may be released.
(185) In a state in which the first plate **100** and the second plate **200** are coupled to each other, the slit **230** may cause a force to act on the second plate **200** in a direction to be separated from the first plate **100**. Accordingly, when the locking of the first coupling portion **120** and the second coupling portion **220** is released, the first plate **100** and the second plate **200** may be easily separated. As a result, disassembly of the first coupling portion **120** and the second coupling portion **220** may be facilitated. The first coupling portion **120** and the second coupling portion **220** may be provided to enable one-touch disassembly.
(186) For example, when the locking of the first coupling portion **120** and the second coupling portion **220** is released, the wing portion **240** formed by the slit **230** may allow the second plate **200** to bounce from the first plate **100**. When locking of the first coupling portion **120** and the second coupling portion **220** is released, the wing portion **240** may be provided to generate a repulsive force. For example, the wing portion **240** may function as a cantilever. For example, when locking of the first coupling portion **120** and the second coupling portion **220** is released, a free end of the wing portion **240** may move downward within a predetermined range.
(187) Meanwhile, in FIGS. **16** to **18**, the first coupling portion **120** of the first plate **100** is illustrated as not including the bridge **123**, but it is not limited thereto. The structure of the bridge **123** shown in FIGS. **6** to **15** may also be applied to the first coupling portion **120** shown in FIGS. **16** to **18**. That is, when the first coupling portion **120** shown in FIGS. **16** to **18** includes the bridge **123**, the first plate **100** and the second plate **200** may be disassembled from with each other by pressing the recessed portion **124** of the bridge **123** in the second direction B. Since this has been described above, a detailed description thereof will be omitted.
(188) FIG. **19** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment. The same reference numerals are assigned to the same configurations as those of the plate assembly described with reference to FIGS. **5** to **15**, and detailed description thereof may be omitted.
(189) Referring to FIG. **19**, the plate assembly **80***a* may further include a cap **400**. The cap **400** may be provided to correspond to the opening **221**. The cap **400** may be detachably mounted on the opening **221**. As the cap **400** is provided to cover the opening **221**, moisture may be prevented from entering the opening **221**. As the cap **400** is provided to cover the opening **221**, introduction of wet air into the opening **221** may be prevented.
(190) For example, the cap **400** may include a cap protrusion **410**. When the cap **400** is mounted on the opening **221**, the cap protrusion **410** may be provided to be locked with the second plate **200**. Although not shown in the drawing, the second plate **200** may include a groove corresponding to the cap protrusion **410**.

(191) FIG. **20** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment. FIG. **21** is a view illustrating a state in which a cap shown in FIG. **20** is mounted on an opening. The same reference numerals are assigned to the same configurations as those of the plate assembly described with reference to FIG. **19**, and detailed description thereof may be omitted.
(192) Referring to FIGS. **20** and **21**, the plate assembly **80***a* may not include a slit **230**. The plate assembly **80***a* may be provided not to include the slit **230** in a portion vulnerable to moisture. For example, when the first plate **100** includes a plurality of first coupling portions **120**, and the second plate **200** includes a plurality of second coupling portions **220** corresponding to the plurality of first coupling portions **120**, the slits **230** may be formed at a surrounding of at least some of the plurality of second coupling portions **220**, and may not be formed at a surrounding of the rest of the plurality of second coupling portions **220**.
(193) The plate assembly **80***a* may include a cap **400** corresponding to the opening **221**. The cap **400** may be detachably mounted on the opening **221**. The cap **400** may be provided to cover the opening **221**. The cap **400** may include a cap protrusion **410** provided to be locked with the second plate **200**.
(194) The plate assembly **80***a* may include a cutout portion **250**. The cutout portion **250** may be formed as a part of the second body **210** that has been cut out. The cutout portion **250** may be disposed adjacent to the opening **221**. For example, in a state in which the cap **400** is mounted on the opening **221**, the operator's hand or tool T may be inserted into the cutout portion **250** to separate the cap **400** from the second plate **200**.
(195) FIG. **22** is an enlarged view illustrating a portion of an example of a plate assembly according to one embodiment. The same reference numerals are assigned to the same configurations as those of the plate assembly described with reference to FIGS. **5** to **15**, and detailed description thereof may be omitted.
(196) Referring to FIG. **22**, the plate assembly **80***a* may further include a cover **500**. The second plate **200** may include the cover **500**. The cover **500** may be provided to correspond to the opening **221**. The cover **500** may be provided to open and close the opening **221**. The cover **500** may be rotatably provided. As the cover **500** is provided to cover the opening **221**, moisture may be prevented from entering the opening **221**. As the cover **500** is provided to cover the opening **221**, introduction of wet air into the opening **221** may be prevented.
(197) The cover **500** may include a cutout portion **510**. The cutout **510** may be formed as a part of the cover **500** that has been cut out. For example, in a state in which the cover **500** closes the opening **221**, the operator's hand or tool T may be inserted into the cutout portion **510** to rotate the cover **500** so that the cover **500** may open the opening **221**.
(198) FIG. **23** is an enlarged view illustrating a portion of an example of a plate assembly according to an embodiment. FIG. **24** is a view illustrating a state in which the cover shown in FIG. **23** covers the opening. The same reference numerals are assigned to the same configurations as those of the plate assembly described with reference to FIG. **22**, and detailed description thereof may be omitted.
(199) Referring to FIGS. **23** and **24**, the plate assembly **80***a* may not include the slit **230**. The plate assembly **80***a* may be provided not to include the slit **230** in a portion vulnerable to moisture. For example, the first plate **100** includes a plurality of first coupling portions **120**, and the second plate **200** includes a plurality of second coupling portions **220** corresponding to the plurality of first coupling portions **120**, the slits **230** may be formed at a surrounding of at least some of the plurality of second coupling portions **220**, and may not be formed at a surrounding of the rest of the plurality of second coupling portions **220**.
(200) The plate assembly **80***a* may include a cover **500** corresponding to the opening **221**. The cover **500** may be provided to open and close the opening **221**. The cover **500** may be provided to cover the opening **221**.

(201) The cover **500** may include a cutout portion **510**. The cutout portion **510** may be formed as a part of the second body **210** that has been cut out. The cutout portion **510** may be disposed adjacent to the opening **221**. For example, in a state in which the cover **500** covers the opening **221**, an operator's hand or tool T may be inserted into the cutout portion **510** to rotate the cover **500** so that the cover **500** may open the opening **221**.
(202) Embodiments of the disclosure may provide a plate assembly including a first plate: and a second plate detachably coupled to the first plate. The first plate includes an opening: a coupling body extending along a direction in which the first plate and the second plate are arranged to form the opening: and a locking portion protruding from the coupling body toward the opening. The second plate may include a protruding portion formed to be inserted into the opening and coupled to the locking portion, the protruding portion including a first coupling protrusion extending in a first direction toward the first plate and a second coupling portion spaced apart from the first coupling protrusion and extending in the first direction: and a bridge connecting the first coupling protrusion and the second coupling protrusion, the bridge forming a recessed portion that is recessed in a second direction opposite to the first direction.
(203) The first plate may further include a slit spaced apart from the opening. The slit may be formed such that a force acts on the first plate in a direction to be separated from the second plate in a state in which the first plate and the second plate are coupled to each other.
(204) As the first coupling protrusion and the second coupling protrusion are pressed to come closer, the locking portion and the protruding portion may be separated from each other.
(205) As the recessed portion is pressed in the second direction, the locking portion and the protruding portion may be separated from each other.
(206) The protruding portion and the bridge may be provided to be elastically deformable.
(207) According to an aspect of the disclosure, the plate assembly and the home appliance may allow for easy coupling and/or disassembling between parts.
(208) According to an aspect of the disclosure, the plate assembly and the home appliance may be improved in assemblability.
(209) According to an aspect of the disclosure, the plate assembly and the home appliance may prevent moisture from entering a coupling region between parts.
(210) According to an aspect of the disclosure, the plate assembly and the home appliance may be improved in aesthetics.
(211) According to an aspect of the disclosure, the plate assembly and the home appliance may be improved in sealing property.
(212) Aspects of embodiments of the disclosure are not limited to those described above, and other aspects that are not described will be clearly understood by those skilled in the art from the description.
(213) Specific embodiments illustrated in the drawings have been described above. However, the disclosure is not limited to the embodiments described above, and those of ordinary skill in the art to which the disclosure pertains may make various changes thereto without departing from the gist of the technical spirit of the disclosure defined in the claims below.

Claims

1. A refrigerator comprising: a storage compartment; an evaporator configured to generate cold air; and a cold air duct configured to guide the cold air generated in the evaporator to the storage compartment, the cold air duct including: a front plate facing the storage compartment, the front plate including: a front body, and a first coupling portion on a rear surface of the front body, a rear plate detachably coupled to a rear of the front plate, the rear plate including: a rear body, a second coupling portion including: an opening in the rear body, and a locking portion, and a slit in the rear body configured so that the second coupling portion is elastically deformable, and a heat insulating

member disposed between the front plate and the rear plate, wherein the first coupling portion is coupled to the second coupling portion as a portion of the first coupling portion is inserted into the opening in the rear body, and the locking portion protrudes toward and contacts the portion of the first coupling portion inserted into the opening in the rear body, and the first coupling portion and the second coupling portion are configured so that the first coupling portion and the second coupling portion are decouplable by an external force pressing on the first coupling portion.

2. The refrigerator of claim 1, wherein the first coupling portion includes: a first coupling protrusion extending in a first direction directed toward the rear plate, a second coupling protrusion spaced apart from the first coupling protrusion and extending in the first direction, and a bridge connecting the first coupling protrusion and the second coupling protrusion, the bridge including a recessed portion recessed in a second direction opposite to the first direction.

3. The refrigerator of claim 2, wherein in response to the external force pressing on the recessed portion of the bridge in the second direction in a state in which the front plate and the rear plate are coupled to each other, the first coupling protrusion and the second coupling protrusion are moved toward one another so that the first coupling portion and the second coupling portion are decoupled.

4. The refrigerator of claim 2, wherein in response to the external force pressing the first coupling protrusion and the second coupling protrusion to be moved toward one another in a state in which the front plate and the rear plate are coupled to each other, the bridge is elastically deformed and the first coupling portion and the second coupling portion are decoupled.

5. The refrigerator of claim 1, further comprising: a cap corresponding to the opening and detachably mounted on the opening to prevent moisture from entering the opening.

6. The refrigerator of claim 1, wherein the rear plate includes a cover configured to open and close the opening to prevent moisture from entering the opening.

7. The refrigerator of claim 2, wherein the slit extends along a direction in which the first coupling protrusion and the second coupling protrusion are arranged.

8. The refrigerator of claim 2, wherein the slit includes: a first slit portion, and a second slit portion spaced apart from the first slit portion, and the first coupling portion is disposed between the first slit portion and the second slit portion in a state in which the front plate and the rear plate are coupled to each other.

9. The refrigerator of claim 2, wherein the second coupling portion includes: a coupling body configured to surround the portion of the first coupling portion inserted into the opening in the rear body, and the slit includes: a first slit forming portion along a portion of an edge of the coupling body, a second slit forming portion extending in a straight line from a first side of the first slit forming portion, and a third slit forming portion extending in a straight line from a second side of the first slit forming portion.

10. The refrigerator of claim 2, wherein the slit includes: a first slit forming portion spaced apart from the opening and extending in a direction perpendicular to a direction in which the first coupling protrusion and the second coupling protrusion are arranged, a second slit forming portion extending from a first side of the first slit forming portion toward the opening along the direction in which the first coupling protrusion and the second coupling protrusion are arranged, and a third slit forming portion extending from a second side of the first slit forming portion toward the opening along the direction in which the first coupling protrusion and the second coupling protrusion are arranged.

11. The refrigerator of claim 2, wherein the first coupling portion includes: a first groove on the first coupling protrusion, and accessible through the opening from behind the rear body, and a second groove on the second coupling protrusion, and accessible through the opening from behind the rear body.

12. The refrigerator of claim 2, wherein the bridge includes: a first bridge portion extending from the first coupling protrusion to the recessed portion, and sloping toward the front body, and a

second bridge portion extending from the second coupling protrusion to the recessed portion, and sloping toward the front body.

13. The refrigerator of claim 12, wherein the recessed portion is provided at a center of the bridge.

14. The refrigerator of claim 1, wherein the heat insulating member is configured to be in surface contact with at least one of the front plate and the rear plate.

15. The refrigerator of claim 1, wherein the heat insulating member includes: a support hole corresponding to the opening of the second coupling portion and configured to accommodate the first coupling portion, and a seating portion extending inwardly from a rear surface of the heat insulating member about the support hole, and configured to support the locking portion of the second coupling portion.